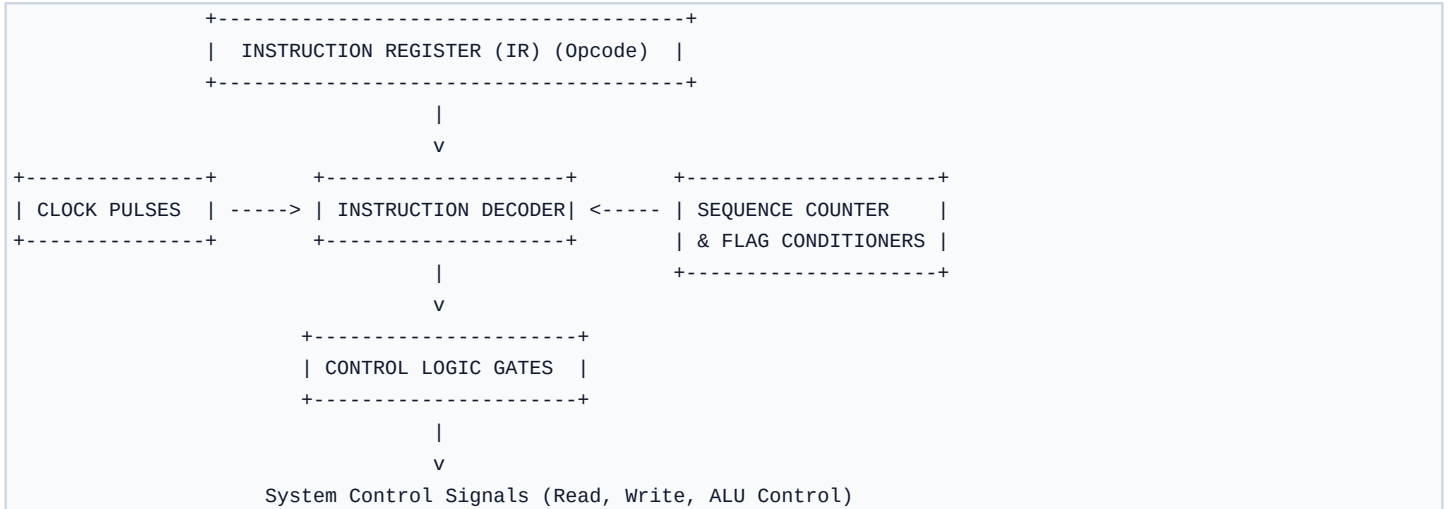


5-Mark Questions & Answers -- MPCA Unit 2: Processor Organization & Architecture

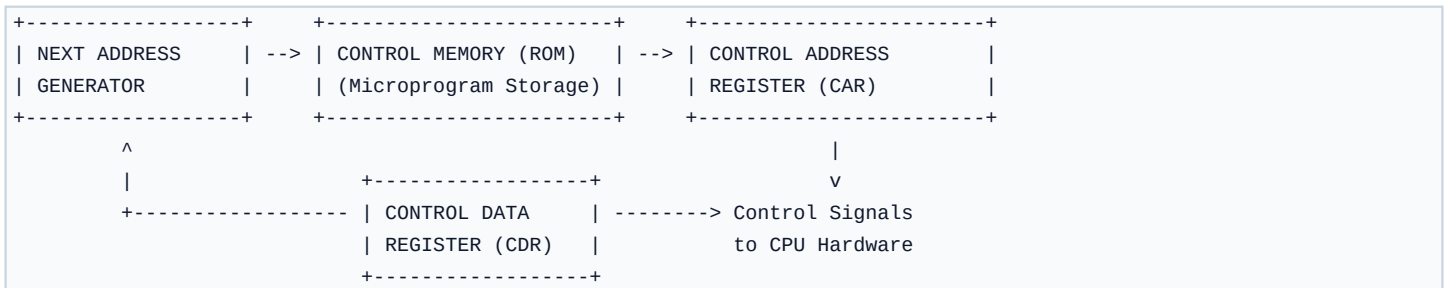
Q1. Explain hardwired control unit design with a block diagram.



1. Opcode Decoder: Decodes machine instruction opcode into active signal lines.

2. Se

Q2. Explain microprogrammed control unit with microinstruction sequencing and execution.



1. Control Memory (ROM): Holds sequences of microinstructions (microprogram) for every machine instruction opcode.

2. Co

Q3. Explain instruction pipelining with a diagram and discuss speedup achieved.

Clock Cycle	:	C1	C2	C3	C4	C5	C6	C7
Instr 1	:	[IF]	[ID]	[EX]	[MEM]	[WB]		
Instr 2	:		[IF]	[ID]	[EX]	[MEM]	[WB]	
Instr 3	:			[IF]	[ID]	[EX]	[MEM]	[WB]

Speedup Calculation

For a k

Maxi

*Ca

- *Ca

2. SIM

2. De

2. De